



Optimal robust scheduling of energy-water nexus system using robust optimization technique

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ABSTRACT

Clean energy and water are two crucial resources that must be delivered safely in every civilization. Water distribution and electricity transmission networks, which provide these precious resources, have traditionally been conceived of as independent, uncoupled systems. However, they are inextricably linked in what is known as the energy-water nexus in reality. This paper considers a nexus system constructed from combined water-power (CWP) and single water and power energy resources to satisfy the power and water demands. Furthermore, a robust optimization technique (ROT) is proposed to model the uncertainties of water and power demands. ROT models the worst realizations of the uncertain parameters. The obtained results are proper strategy for the conservative system operators. The robust-based optimization problem is solved using OQNLP solver under the GAMS software. Results state that by considering the proposed robust-based optimization model, the total operation cost will be increased by about 9.52% for the conservative operators.

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1. Introduction

1.1. motivation

Recently, lots of studies came to attention in the combined energy resources section (Fanwei Meng et al., 2018). As a result, strict and helpful policies have been adopted in several nations (Peng et al., 2021). This combined energy nexus system involves two energy resources like water and power energies (Wang et al., 2019). Nexus system has some prerequisites for every specific country, which differs from city to city according to the infrastructure and geographical locations. Using this technology, particularly in the famine and drought regions, will be a promising way to fulfill the water and power demands (Duan et al., 2021). Energy resources consumption soaring, various policies of nations, and as a more critical factor world population ascending, made the politicians come up with a solution to meet the food, water, and power demands (Kermani et al., 2017). On the other hand, the lack of water resources made a culprit issue for electric energy generation.

Therefore, cultural steps and strong policies should be taken to account in both national and international terms.

Moreover, energy systems optimization problems are faced with uncertainties mainly in demands, prices, and renewable generations (Jiang et al., 2021). The significant effects of this uncertain parameter on the optimization problems results have led to investigation by researchers using different methods (Zhang et al., 2021). Also, the risk assessment is involved in the uncertainty management methods for help to optimal decision making under uncertainty. The robust optimization approach is one of the uncertainty modeling methods that does not need to large number of historical data. decision making with robust optimization only required to a deviation bounds of uncertain parameters. Besides, due to significantly affecting the nexus systems in the agriculture and energy sectors and changing the demands of this system during the operation horizon, the uncertainty modeling approach becomes necessary.

However, based on the description represented above, the nexus system may face uncertainties that change the decision-making in these systems. The different arrangements of nexus systems are represented to introduce the various nexus systems and the used modeling approaches. By introducing the different arrangements of the nexus system and mathematical models, the different uncertainty modeling methods are reviewed to clarify the

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Nomenclature

Indices

g	Thermal generators
w	Water desalination unit
c	CWP plants
t	Time

Parameters and variables

C_p	Fuel cost of thermal generators (\$/MW)
C_w	Fuel cost of water desalination unit (\$/MW)
C_{cwp}	Fuel cost of CWP plants (\$/MW)
$P_{g,t}$	Generated power from thermal generators units (MW)
$W_{w,t}$	Produced clean water from water desalination units (m^3/h)
$P_{c,t}$	Production power from CWP units (MW)
$W_{c,t}$	Clean water generation of CWP units (m^3/h)
a_g, b_g, c_g	Fuel cost factors of thermal generators (\$/MW ² , \$/MW, \$)
a_w, b_w, c_w	Fuel cost factors of water desalination units (\$/MW ² , \$/MW, \$)
$\alpha_c, \beta_c, \gamma_c, \lambda_c, \xi_c, \zeta_c$	Fuel cost factors of CWP plants (\$/MW ² , \$/MW, m^3/h , m^3/h , m^6 , \$/MW, \$/MW, \$/MW, \$)
$U_{g,t}$	A binary variable for thermal generators which is one if generators are on otherwise 0
$U_{w,t}$	A binary variable for desalination units which is one if units are on otherwise 0
$U_{c,t}$	A binary variable for CWP producers which is one if units are on, otherwise 0
q, y, z	Dual and auxiliary variables of the ROT
E_t	Electrical power demand (MW)
W_t	Clean water demand (m^3/h)
$p_g^{\min}$	Minimum power production of thermal power plants (MW)
$p_c^{\min}$	Minimum power production of CWP units (MW)
$p_g^{\max}$	Maximum power production of thermal power plants (MW)
$p_c^{\max}$	Maximum power production of CWP units (MW)
$W_w^{\min}$	Minimum clean produced water of desalination units (m^3/h)
$W_c^{\min}$	Minimum clean produced water of CWP units (m^3/h)
$W_w^{\max}$	Maximum clean produced water of desalination units (m^3/h)
$W_c^{\max}$	Maximum clean produced water of CWP units (m^3/h)
$R_c^{\min}$	Minimum power-water ratio of CWP unit (MWh/ m^3)
$R_c^{\max}$	Maximum power-water ratio of CWP unit (MWh/ m^3)
DR_g	Ramp down of thermal power plant
UR_g	Ramp up of thermal power plant
Γ	Budget of uncertainty

disadvantages of these methods to highlight the application of the robust optimization method.

1.2. Literature review

In (Chen et al., 2019), inter-regional trade is proposed for a city-scale framework to investigate direct and embodied energies such

as water and power resources. A review paper researched the energy, water, and carbon dioxide nexus system and investigated it from every aspect. Finally, it introduced hints for the nexus system (Fanxin Meng et al., 2019). Possible ways of managing the nexus system through three integrated management strategies (INRM, IWRM, ISWM) are investigated in (Roidt & Avellán, 2019). A detailed understanding of the analysis of the nexus system is proposed, and the main problems are investigated to obtain a better result from these systems in (Cabello et al., 2019). According to (Namany et al., 2019b), a multi-objective stochastic optimization approach is used to realize the optimum way of satisfying water and power energies and obtain a sustainable system in the trade-off market. In (Guan et al., 2019), systematic research is adopted to analyze the water and power demand energy resources by considering land resources as a national project in China. To achieve an inter-connected and robust approach in the nexus system and considering agricultural aspects, computational scaffolding of a framework is proposed in (Govindan & Al-Ansari, 2019). In (Namany et al., 2019a), agent-based modeling and game theory methods are used for dynamic decision-making to attract the attention of policymakers. In (Xu et al., 2019), different types of intertwined virtual water and power resources are investigated to fulfill the missing knowledge by considering China as a case study. In (Tsolas et al., 2019), an organized approach is proposed to design and optimize the nexus network using a graph-theoretic procedure under the two different cases of minimum and maximum generation of both water and power energies. In (Biggs et al., 2015), the dependence of human life and nexus approaches and its relation with the world's population rise is analyzed. An integrated method that identifies the intertwined nexus systems with the food section is investigated on a household scale, and as a result, parameters such as family size, amount of income, the diet had a significant impact on these systems (Hussien et al., 2017). In (Obringer et al., 2019), a multi-response framework is proposed to the water-power demand nexus system by considering climate change.

In (Jabari et al., 2019), a day-ahead optimization procedure was applied to a water-heat-power system by considering all technical constraints to minimize the fuel cost of the water desalination unit. A comprehensive review of desalination units integrated with renewable energy resources alongside essential factors such as base technology, cost of operation, and environmental issues are investigated (Alkaisi et al., 2017). In (Hicks et al., 1989), all the required principles of designing, operation, and citing of desalination units are reviewed. In (Shahzad et al., 2017), a state of art review under the intertwined energy resources is expressed to save energy and investigate environmental issues. Cogeneration plants, which produce both water and electricity at the same time, are common in the Gulf Region. They've been shown to be more thermodynamically successful and economically viable than only one power and water facilities (Hamed et al., 2006). A water-power nexus system combined with air-heated humidification dehumidification (HDH) desalination unit is proposed in (He et al., 2017), to satisfy the water and power demands. In (He et al., 2018), for specific values of water production and gained-output ratio, a system is introduced to take advantage of the cost for water production from the desalination unit. In (Arani et al., 2017), a strategy for analyzing the economic issues of thermal systems is proposed; other units such as combined gas turbine and desalination units are investigated. Intertwined water and power energy resources are investigated for a city through a system-based framework in (Chen & Chen, 2016). In the stated research, consumed energy and water consumption are merged into unified hybrid energy. These units are compared separately to obtain effective results.

The water and power uncertainties have significant effects on the optimal operation of the nexus system. The impacts of the water uncertainty are investigated in (Guieysse et al., 2013). The wa-

ter and power demand uncertainty in the nexus systems is modeled using the chance-constrained modeling approach based on scenario method in (Stuhlmacher & Mathieu, 2020) to determine the optimal schedule of water pumps considering the forecasting errors of demands. Furthermore, in terms of the uncertainty modeling in the nexus systems, ref (Ogbolumani & Nwulu, 2021) uses the stochastic fuzzy-based approach, and (Namany et al., 2018) proposed a non-cooperative game. Moreover, in (Tan et al., 2021) a two-stage stochastic optimization is also used in uncertainty modeling in the water and power planning problem. To model the uncertainties in a energy-water nexus, the uncertain parameters are expressed via the interval values and fuzzy random variables in (Lv et al., 2018). An energy-water nexus system planning problem, for low-carbon system design, is investigated in (Ji et al., 2020) according to needs for a robust solutions against the uncertainties considering a trade off between cost and risk. Moreover, the importance of the water demand uncertainty modeling and used methods is investigated in (Tiwari & Adamowski, 2013), and an ensemble wavelet-bootstrap-neural network model is proposed for the forecasting process.

1.3. Novelty and contribution

Based on the reviewed papers, the modeling of the uncertainty in the water and power nexus system has not been investigated decently. Moreover, according to the reviewed papers, the water and power demand's uncertainties have considerable impacts on the nexus systems. Thus, in this paper, the power and water demand's uncertainty is considered a big challenge of the nexus systems. Modeling the uncertainties using robust optimization is the contribution of this paper. To deal with the uncertainty of nexus systems, which has a great impact on the water and power demands, using one of the optimization approaches will be a promising way to handle the uncertainty of these systems. In this paper, the robust optimization technique (ROT) will be used for optimizing the water and power demands for the proposed nexus system. Also, a deterministic approach will be applied to the nexus system to point out the difference between the ROT and deterministic approaches. So, the arrangement of the proposed paper will be as follows:

- Applying the robust optimization technique to model the uncertainties of the water and power demands in the nexus system
- Proposing a robust strategy versus deterministic strategy

Furthermore, the conceptual illustration of the proposed system is represented in Fig. 1.

Section 1 presented the introduction and dealt with the overview of the related surveys. Section 2 will be dedicated to problem formulation and its explanation. The general formulation of ROT and applied ROT to the base problem are presented in section 3. The case study is expressed in section 4. Finally, section 5 intended to point out the conclusion of the paper.

2. Deterministic based problem formulation

The proposed objective function aims to minimize the total cost of all production units. To clarify the issue, production units are composed of water, power, and CWP plants. The priority of the proposed paper is to optimize the total cost of these production units by using the ROT procedure. So, equation (1) expresses the objective function as follows:

$$\text{Objective Function} = C_p + C_w + C_{cwp} \quad (1)$$

To investigate the objective function in detail, three main parts of the objective function are expressed through equations (2)–

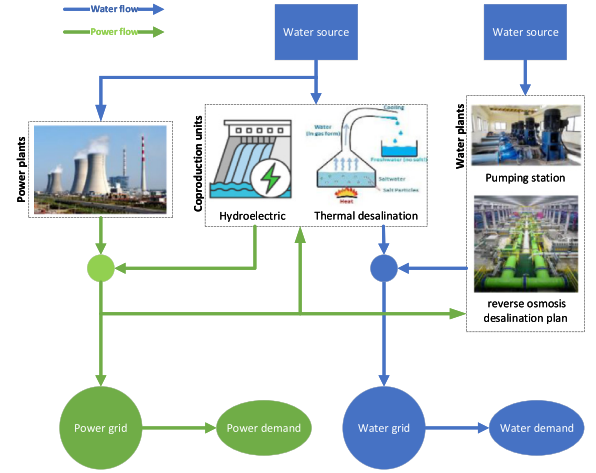


Fig. 1. Overview of the proposed system

(4). Equation (2) expresses the cost of all power production thermal units, which is a quadratic function of the generated power by thermal units. Also, three fuel cost coefficients of power generation are influenced by three indicated coefficients. Similarly, equations (3) and (4) present the cost of desalination units' water production and CWP plants' water production, respectively.

$$C_p = \sum_{g,t} (a_g P_{g,t}^2 + b_g P_{g,t} + c_g U_{g,t}) \quad ; \quad \forall g \in \Omega_p \quad \forall t \in T \quad (2)$$

$$C_w = \sum_{w,t} (a_w W_{w,t}^2 + b_w W_{w,t} + c_w U_{w,t}) \quad ; \quad \forall w \in \Omega_w \quad \forall t \in T \quad (3)$$

$$C_{cwp} = \sum_{c,t} (\alpha_c P_{c,t}^2 + \beta_c P_{c,t} W_{c,t} + \gamma_c W_{c,t}^2 + \lambda_c P_{c,t} + \xi_c W_{c,t} + \zeta_c U_{c,t}) \quad ; \quad \forall c \in \Omega_{cwp}, \quad \forall t \in T \quad (4)$$

In (2)–(4), the cost function coefficients are properly sized positive constant matrices based on their individual manufacturing units' heat rate characteristics. In addition, the cost functions C_p , C_w , C_{cwp} are considered as quadratic functions with respect to their production variables. For all three types of generation units, this is a generally held notion in which details of these functions are given in (Wood et al., 2009).

Moreover, equations (5) and (6) are used to obtain plausible results to meet the balance constraint in the water and power production units. These equations mean that all the generating units both for water and power should meet the water and power demands.

$$\sum_{g \in \Omega_p} P_{g,t} + \sum_{c \in \Omega_{cwp}} P_{c,t} = E_t \quad ; \quad \forall t = 1, 2, 3, \dots, T \quad (5)$$

$$\sum_{c \in \Omega_{cwp}} W_{c,t} + \sum_{w \in \Omega_w} W_{w,t} = W_t \quad ; \quad \forall t = 1, 2, 3, \dots, T \quad (6)$$

In the following, the main boundaries of the proposed problem are presented. Equation (7) – (8) pertains to the power production limit generated from thermal and CWP units, respectively. Therefore, power generation will have a specific value of production due to the thermal and CWP plant limits.

$$P_g^{\min} \times U_{g,t} \leq P_{g,t} \leq P_g^{\max} \times U_{g,t} \quad ; \quad \forall g \in \Omega_p \quad \forall t \in T \quad (7)$$

$$P_c^{\min} \times U_{c,t} \leq P_{c,t} \leq P_c^{\max} \times U_{c,t} \quad ; \quad \forall c \in \Omega_{cwp} \quad \forall t \in T \quad (8)$$

Similar to the above equation's concept, equations (9)–(10) are related to the total water production of water desalination units

and CWP plants, respectively.

$$W_w^{\min} \times U_{w,t} \leq W_{w,t} \leq W_w^{\max} \times U_{w,t}; \quad \forall w \in \Omega_w \quad \forall t \in T \quad (9)$$

$$W_c^{\min} \times U_{c,t} \leq W_{c,t} \leq W_c^{\max} \times U_{c,t}; \quad \forall c \in \Omega_{cwp} \quad \forall t \in T \quad (10)$$

According to equation (11), the water and power generation ratio is bounded through their minimum and maximum water-power production, which relates to the conditions of the CWP plants.

$$R_c^{\min} \leq \frac{P_{c,t}}{W_{c,t}} \leq R_c^{\max}; \quad \forall g \in \Omega_p \quad \forall t \in T \quad (11)$$

Equation (12) indicates the ramp-up and ramp-down values for thermal power plants, which depend on previous power.

$$\begin{aligned} \text{Max}(P_g^{\min} P_g^{t-1} - DR_g) &\leq P_{g,t} \\ &\leq \text{Min}(P_g^{\max} P_g^{t-1} + UR_g); \quad \forall g \in \Omega_p \quad \forall t \in T \end{aligned} \quad (12)$$

3. ROT based problem formulation

The robust optimization approach is a well-known uncertainty modeling approach in optimization problems. Unlike the stochastic-based methods, implementing this approach doesn't need historical data from uncertain parameters, which is advantageous for the ROT. Hence, this approach can be modeled in the optimization problems by defining a deviation range for the uncertain parameters. Thus, considering a proper deviation bound for the uncertain parameter would be a disadvantage for the ROT. Accordingly, the results of the robust optimization give the conservation results against the deviation of the uncertain parameters. This section will introduce the general form of robust optimization technique (Dominguez et al., 2012) as follows:

$$\text{Min} \sum_{j=1}^n c_j \cdot x_j \quad (13)$$

Subject to:

$$\sum_{j=1}^n a_{ij} \cdot x_j \leq b_i \quad i = 1, 2, \dots, m \quad (14)$$

$$x_j \geq 0 \quad j = 1, 2, \dots, n \quad (15)$$

$$x_j \in \{0, 1\} \text{ for some } j = 1, 2, \dots, n \quad (16)$$

In the above equations, x_j stands for optimization variables, a_{ij} and b_i are expressed for factors of the matrix constraints and coefficient of right-hand-side vector, respectively. Values a_{ij} are independent, symmetrical, and have a random variable. It should be noted that these values are limited with an unknown distribution function but with a specific domain $[a_{ij} - \hat{a}_{ij} \quad a_{ij} + \hat{a}_{ij}]$. For values $\hat{a}_{ij} \geq 0$, deviation from the average values a_{ij} will be $\hat{a}_{ij}$. Also, in the case that uncertainty exists, a control parameter Γ_i will be used to formulate the specific problems to regulate the robustness of the considered technique versus the expected value of conservatism in the solution. This control parameter Γ_i will be in the boundary of $[0, |J_i|]$ where $J_i = \{j | \hat{a}_{ij} \geq 0\}$. It is noteworthy to state that Γ_i can be a non-integer value between $[0, 1]$. More details about the modeling of the robust optimization methodology can be found in reference (Nourollahi, Salyani, et al., 2021) and (Nourollahi, Mohammadi-Ivatloo, et al., 2021). Moreover, the mathematical details of the ROT to understanding the auxiliary variables and hyperparameters like Gamma are described in (Bertsimas & Sim, 2004).

If there is an uncertain parameter which includes b_i in the proposed problem, an ancillary component so-called x_{n+1} will participate in the above equations, and a new form of the equation will be as follows:

$$\text{Minimize}_{x_j} \sum_{j=1}^n c_j \cdot x_j \quad (17)$$

Subject to:

$$\sum_{j=1}^n a_{ij} \cdot x_j - b_i \cdot x_{n+1} \leq 0 \quad i = 1, 2, \dots, m \quad (18)$$

$$x_j \geq 0 \quad j = 1, 2, \dots, n \quad (19)$$

$$x_{n+1} = 1 \quad (20)$$

$$x_j \in \{0, 1\} \text{ for some } j = 1, 2, \dots, n \quad (21)$$

In the updated formulas, b_i stands as a random variable and $\hat{b}_i$ can only take values that are in the region of $[b_i - \hat{b}_i \quad b_i + \hat{b}_i]$ where $\hat{b}_i \geq 0$, and deviation from the average values of b_i coefficients is $\hat{b}_i$. Finally, the resultant and general form of the ROT will be as follows:

Minimize

$$\sum_{j=1}^n c_j \cdot x_j \quad (22)$$

Subject to:

$$\sum_{j=1}^n a_{ij} \cdot x_j - b_i + z_i \cdot \Gamma_i + q_{i,n+1} \leq 0 \quad i = 1, 2, \dots, m \quad (23)$$

$$z_i + q_{i,n+1} \geq \hat{b}_i \cdot y_{n+1} \quad i = 1, 2, \dots, m \quad (24)$$

$$q_{i,n+1} \geq 0 \quad i = 1, 2, \dots, m \quad (25)$$

$$y_{n+1} \geq 0 \quad (26)$$

$$z_i \geq 0 \quad i = 1, \dots, m \quad (27)$$

$$y_{n+1} \geq 1 \quad (28)$$

$$x_j \geq 0 \quad j = 1, \dots, n \quad (29)$$

$$x_j \in \{0, 1\} \text{ for some } j = 1, \dots, n \quad (30)$$

In the above equation, $q_{i,n+1}$, y_{n+1} and z_i are dual variables of the proposed problem, and there is another auxiliary component y_{n+1} that is defined to obtain linear terms.

3.1. Applied ROT to the base problem

According to the general procedure of ROT, which is expanded in the previous section, objective function will remain constant to apply ROT to the base problem. It should be noted that in this proposed problem, equations (5) and (6) should be updated through equations (33)–(38) and (39)–(44), respectively, to participate in the ROT.

$$\text{Objective Function} = C_p + C_w + C_{cwp} \quad (31)$$

$$\text{constraints: (2) – (4) and (7) – (12)} \quad (32)$$

Table 1
Specifications of installed units

	Power units				Coproduction units			water
	unit #1	unit #2	unit #3	unit #4	unit #1	unit #2	unit #3	unit #1
Maximum power capacity (MW)	500	400	400	350	800	600	400	0
Maximum water capacity (m ³ /h)	0	0	0	0	200	150	100	250
Minimum power capacity (MW)	0	0	0	0	160	120	80	0
Minimum water capacity (m ³ /h)	0	0	0	0	30	23	15	0
Minimum product ration (MWh/m ³)	-	-	-	-	4	4	4	-
Maximum product ration (MWh/m ³)	-	-	-	-	9	9	9	-

$$\sum_{g \in \Omega_p} P_{g,t} + \sum_{c \in \Omega_{cwp}} P_{c,t} = \tilde{E}_t + Z_1(t).Gama(t) + Q_1(t); \quad \forall t = 1, 2, 3, \dots, T \quad (33)$$

$$Z_1(t) + Q_1(t) \geq DE_{\max}(t) \times Y_1(t) \quad (34)$$

$$Q_1(t) \geq 0 \quad (35)$$

$$Y_1(t) \geq 1 \quad (36)$$

$$Z_1(t) \geq 0 \quad (37)$$

$$Y_1(t) \geq 0 \quad (38)$$

Similarly, the following equations will be applied to the second uncertain parameter, which [equation \(6\)](#) has it already.

$$\sum_{c \in \Omega_{cwp}} W_{c,t} + \sum_{w \in \Omega_{cwp}} W_{w,t} = \tilde{W}_t + Z_2(t).Gama(t) + Q_2(t); \quad \forall t = 1, 2, 3, \dots, T \quad (39)$$

$$Z_2(t) + Q_2(t) \geq DW_{\max}(t) \times Y_2(t) \quad (40)$$

$$Q_2(t) \geq 0 \quad (41)$$

$$Y_2(t) \geq 1 \quad (42)$$

$$Z_2(t) \geq 0 \quad (43)$$

$$Y_2(t) \geq 0 \quad (44)$$

In the above equations, the terms $Z_1(t)$, $Q_1(t)$, $Y_1(t)$ are the auxiliary variables and $\tilde{E}_t$ Gama, $DE_{\max}(t)$ are parameters for the electrical power demand, which are considered to perform the ROT. Correspondingly, $Z_2(t)$, $Q_2(t)$, $Y_2(t)$ are the auxiliary variables and $\tilde{W}_t$ Gama, $DW_{\max}(t)$ are the parameters for the water demand.

4. Case study

This study considers a multi-production unit such as water, power, and CWP units to satisfy the water and power demands. Therefore, one water desalination plant, four thermal plants, and three CWP units are considered. To be precise, input data are derived from ([Santhosh et al., 2014](#)). However, the specifications of the installed units are represented in [Table 1](#) to clarify which resources are considered in the proposed model. The cost coefficients of the installed units is represented in [Table 2](#). The represented case study models is modeled a 24 hour operation of the proposed

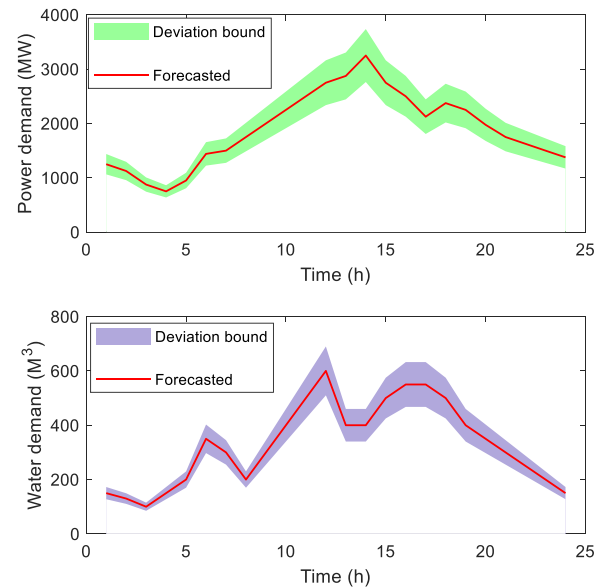


Fig. 2. Power and water demands profil

nexus system. Further, the water and power demands along with the deviation bounds in RO are represented in [Fig. 2](#).

In this study, two cases are investigated to show how this applied optimization approach is practical for uncertainty. These cases are expressed as follows:

Case1: Without applying ROT which is known as the deterministic approach

Case2: With applying ROT

Before discussion on the obtained results, a summary of model statistic would be useful. According to the solve report, there are 604 single variables and 192 integer variables inside 1,060 equations. Besides, there are 264 nonlinear equations in the proposed model which is shown the high computational burden of the proposed model.

From now on, obtained results for both of the cases will be presented. According to the important effects of the uncertain parameters on the optimal results of the optimization problem, the obtained results of the problem will be differed by uncertainty modeling. As mentioned, the robust optimization models the worst realizations of the uncertain parameters that are the conservative operators' strategy. In addition, modeling the worst realization of uncertain parameters will be guaranteed the adequacy of the resources against the worst conditions in the power and water demands. Thus the results are represented in two cases, including the risk-averse and risk-neutral strategies.

The total robust operation cost of all production units is exhibited in [Fig. 3](#). In this proposed problem, ten steps of 0.1 are considered to fulfill the whole Gama in the magnitude of 1. It is evident that, by the increase of Gama, total robust operation cost

Table 2
Cost coefficients of units

	Parameters	Unit 1	Unit 2	Unit 3	Unit 4
Power plants' cost coefficients	A_p	2.069e-4	3.232e-4	1.065e-3	4.222e-4
	B_p	-1.483e-1	-1.854e-1	-6.026e-1	-2.119e-1
	C_p	5.711e+1	5.711e+1	1.268e+2	5.711e+1
Coproduction units' costs coefficients	A_{c11}	7.881e-4	1.773e-3		
	A_{c12}	6.305e-3	1.419e-2		
	A_{c22}	1.261e-2	2.837e-2		
	B_{c1}	-1.475	-2.213		
	B_{c2}	-5.901	-8.851		
	C_c	7.374e+2	7.374e+2		
Water plants' cost coefficients	A_w	1.82E-02			
	B_w	-7.081			
	C_w	7.374			

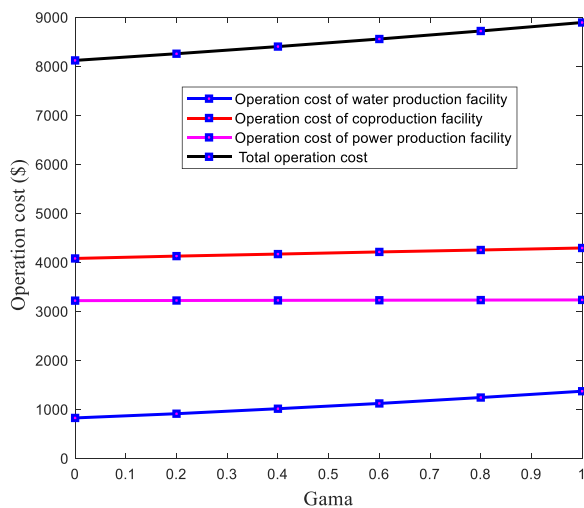


Fig. 3. Operation costs of units versus variation of Γ

risers linearly. To be more precise, the deterministic approach can be obtained by $\Gamma = 0$, which can be read from the obtained curve. Also, $\Gamma = 1$ indicates the ROT approach. For $\Gamma = 0$, the corresponding value of the total operation (the robust cost) is \$ 8,118.8. For $\Gamma = 1$, this value is \$ 8,891.9. By comparing these first and last iterations, it can be seen that there is a sharp increase in the total operation robust cost, which is 9.52% in comparison with the deterministic approach. This result indicates that the ROT's impact on the proposed problem is significant.

According to Fig. 3, the linear function of the robust operation cost of power production units can be seen easily. The more Γ increases, the more robust operation cost of power production increases. To investigate more accurately, for $\Gamma = 0$, which refers to the deterministic approach, the related value of power production cost is \$ 3,217.1. For $\Gamma = 1$ which stands for the ROT, this value increased to \$ 3,231.8. The deviation of power production cost is not remarkable because the increased percentage is 0.46.

As illustrated in Fig. 3, similarly, there is a linear increase in the robust operation cost of the water desalination unit by the increase of its correspondent Γ . To give a clear viewpoint, similarly to the previous explanations, the corresponded value of water desalination cost is \$ 823.8 for $\Gamma = 0$ (deterministic), in which this value rises to \$ 1,368.9 for $\Gamma = 1$ (ROT). By sizing up the calculation, there is a 66% increase in the total operation robust cost of water desalination, which has a significant increase, and it is notable.

By the cost comparison of the 3 it can be concluded that the water facility takes a considerable effect from the water and power uncertainties. These high impacts are illustrated in the first and last Γ 's of Fig. 3, where the operation cost of the water facility is increased significantly against the increasing Γ . Moreover, as it is presented in Fig. 6, the slope of the robust operation cost of the coproduction units is positive and increases as correspondent Γ increases. According to the obtained following values, there is a 5.23% increase in the total operation robust cost of CWP production; for $\Gamma = 0$ obtained value is \$ 4,077.9, and for $\Gamma = 1$ this value is \$ 4,291.2.

Power demand is fully satisfied in both deterministic and robust approaches, proving the correct implementation of the proposed problem. In general, power demand and its production units are increased totally in the ROT case compared to a deterministic one, which is illustrated in Fig. 4. To be more precise, the exact values of power demand and power production of all facilities will be discussed as follows. To point out significant deviations in the deterministic and the ROT approaches, just remarkable deviations are expressed briefly. The power production of thermal plants and CWP units is considered first. Power production of power plant one is increased from 452.78 MW (in the deterministic case) to 500.00 MW in (the ROT case) in hour 13, representing a 10.43% increment. Also, power plant two had some deviation in hour 13, from 347.25 MW (deterministic) to 380.41 MW (ROT). For power plant three, despite previous power plants one and two, there are two deviations in the power production of this unit, which are increased from 301.25 MW to 311.32 MW in hour 13, and from 351.56 MW to 397.88 MW in hour 14. Similar to power plants one and two, the corresponded value for the deterministic model is 297.20 MW, which increases to 322.60 MW in the ROT approach in hour 13. From now on, the power production of CWP units will be discussed. There are both increments and decrements in the power generation of coproduction units in these units, which we will express some of them. For cogeneration plant one, power production decreased from 796.61 MW (in the deterministic case) to 623.90 MW (in the ROT case), which represents a decrement in hour 23. On the contrary, in hour 14, power production is increased from 706.25 MW (deterministic) to 797.12 MW (ROT), which expresses a 12% increment. For the cogeneration plant two, power production is increased from 468.28 MW (in the deterministic case) to 522.51 MW (in the ROT case) in hour 22, and also, it is decreased from 598.65 MW (deterministic) to 467.93 MW (ROT) in hour 24. For the last cogeneration plant three of the power production, there are minor deviations except for two major ones, which are risen from 312.66 MW (in the deterministic case) to 400.00 MW (in the ROT case) in hour 22 and decreased from 400.00 MW (deterministic) to 312.66 MW (ROT) in hour 21.

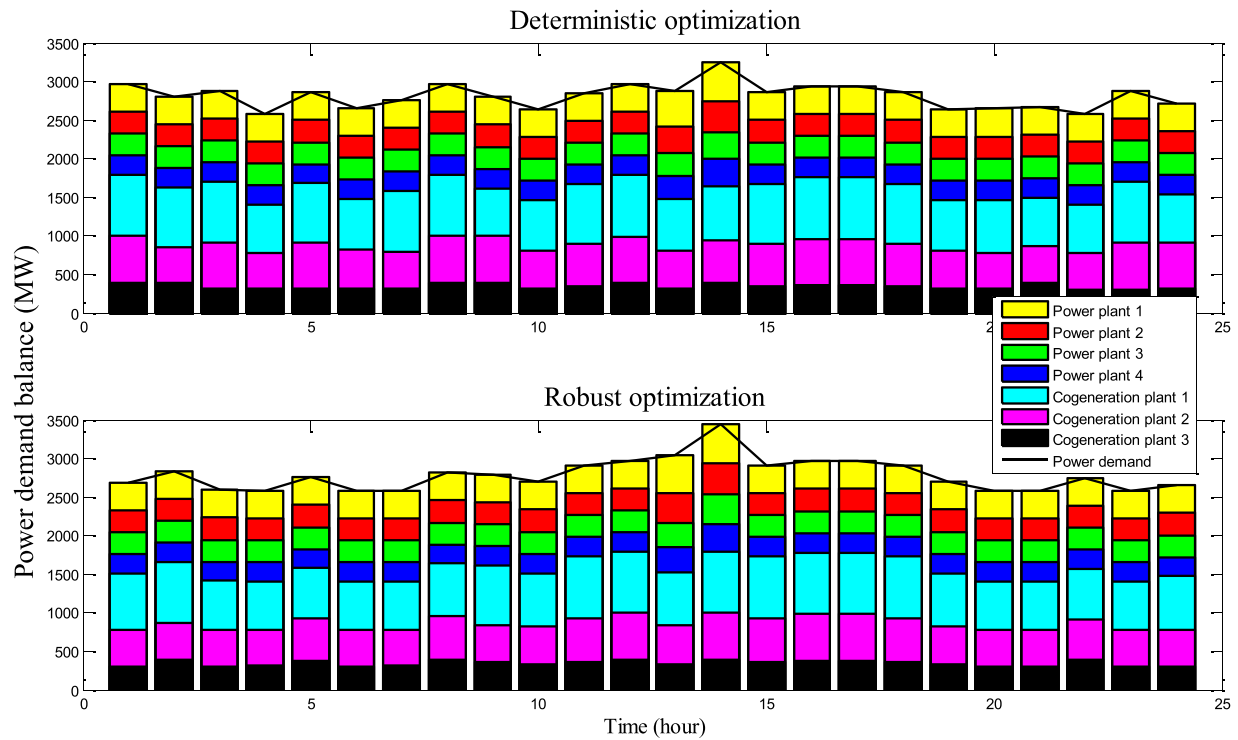


Fig. 4. Applied deterministic and robust approaches in case of power demand balance

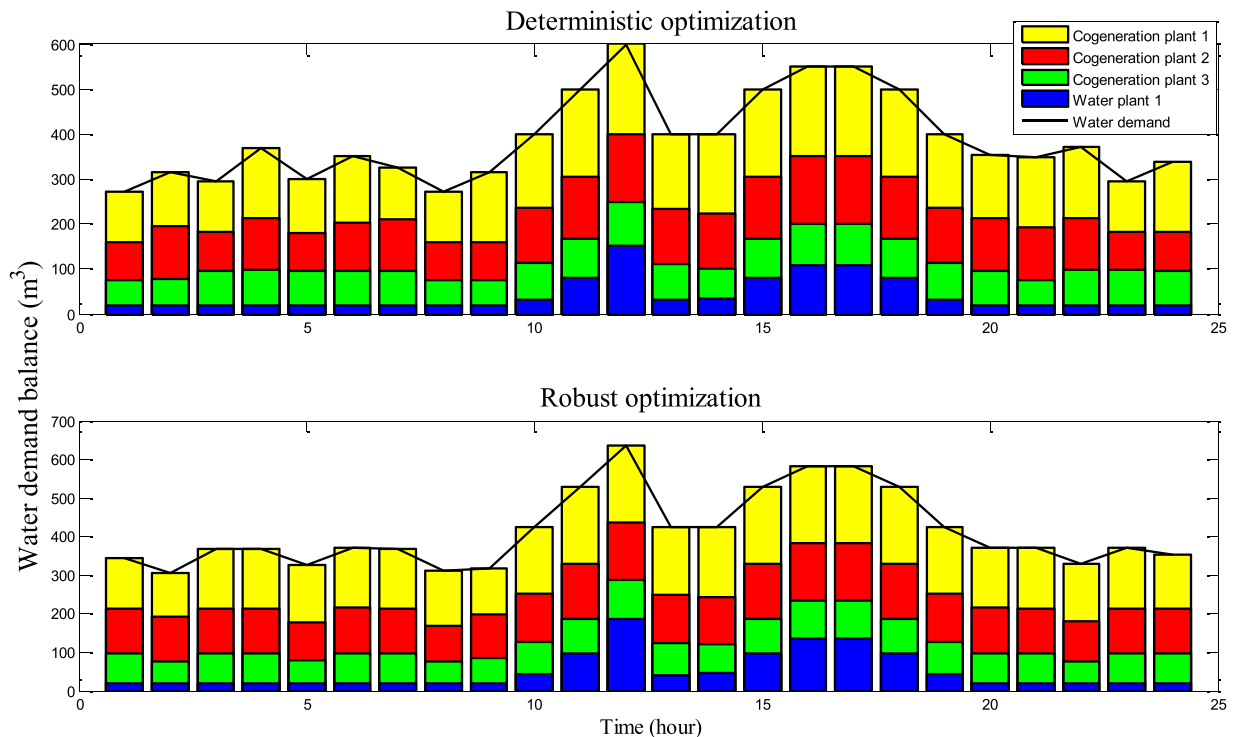


Fig. 5. Applied deterministic and robust approaches in case of water demand balance

According to Fig. 5 and similar to the power demand balance, water demand is met entirely by the water and CWP units. Water demand growth and, as a consequence, water production increment in both the deterministic and robust cases cannot be neglected. To straighten out the obtained figure's main concept and give a better overview as the previous explanation, some of the specific values of water production units will be represented. Wa-

ter production of water plant one varies almost from hour 10 to 19. To be more accurate, for example, in hour 12, water production increased from 150.96 (in the deterministic case) to 186.00 (in the ROT case), which indicates the increment in water production due to ROT application. Now cogeneration units of water production will be analyzed in detail. For cogeneration plant one, both increment and decrement water production can be seen clearly. Wa-

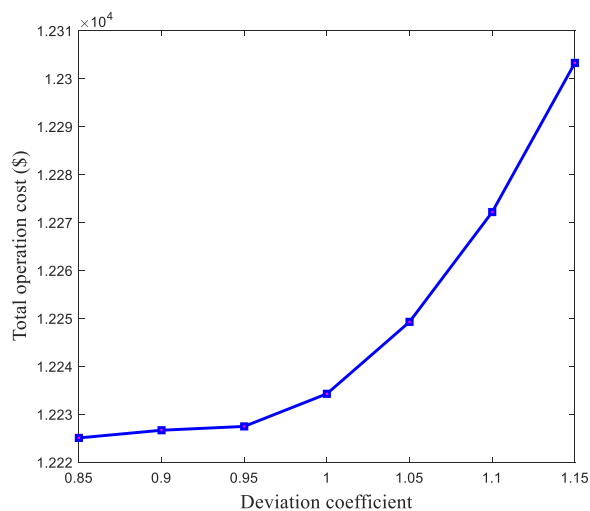


Fig. 6. Cost variation versus different demand coefficients

ter production increased from 113.06 (in the deterministic case) to 155.98 (in the ROT case) in hour 3. Also, this water production decreased from 155.99 (in the deterministic case) to 119.23 (in the ROT case) in hour 9. Furthermore, for cogeneration plant two, water production again increased from 138.58 (in the deterministic case) to 144.35 (in the ROT case) in hour 18. For cogeneration plant three, the highest deviation is related to hour 1, which is increased from 55.96 (in the deterministic case) to 77.95 (in the ROT case).

Water and power demands' increasing and decreasing characteristics are presented in Fig. 4 and Fig. 5, respectively. For example, in power demand due to fluctuations of power production units' production and by considering power demand variation, power demand increased from 2875.00 (in the deterministic case) to 3047.50 (in the ROT case). Similar to the power demand concept, here in hour 1 in the water demand, water production increased from 272.97 (in the deterministic case) to 343.41 (in the ROT case).

4.1. Sensitivity analysis of demands

In this section, a sensitivity analysis is carried on the demands to show the effectiveness of the proposed methodology. The results of this analysis are represented in Fig. 6. According to Fig. 6, increasing the power and water demands will lead to increasing the system's operation cost. Further, Fig. 6 shows that increasing the demands from the nominal value increases the total cost compared to the lower values.

5. Conclusion

This study deals with the uncertainty of water and power demand in a nexus system. This uncertainty is met by taking advantage of the optimization approach, ROT. The proposed problem is solved through two cases, which were deterministic and the ROT approaches. As a result, there is 9.52% increment in the total operation robust cost due to implementing the ROT approach by considering the following values; for $Gama = 0$ which stands for deterministic approach, obtained total operation robust cost is \$ 8,118.8 and for $Gama = 1$ which refers to the ROT approach, the total correspondent operation robust cost is \$ 8,891.9. Due to the peak periods and conditions of the producing plants, main produced power, water, or cogeneration plants were taking place to satisfy the power and water demands in both deterministic and ROT cases. As a result, all the water and power demands have been satisfied through water and power-producing units.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Qun Guo: Data curation, Methodology, Writing – original draft. **Tiantong Guo:** Formal analysis, Software, Writing – original draft. **Qiannan Tian:** Validation, Visualization, Writing – original draft. **Sayyad Nojavan:** Conceptualization, Investigation, Supervision, Writing – review & editing.

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